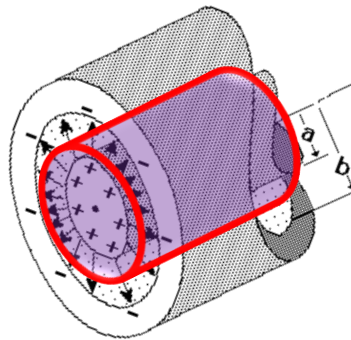
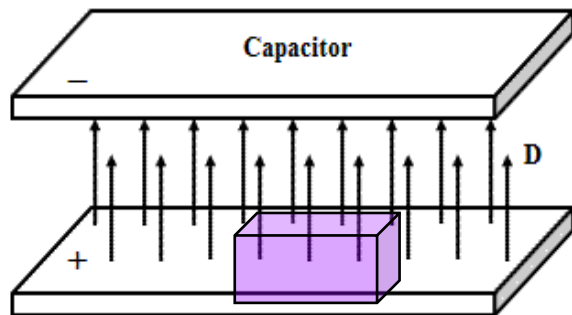
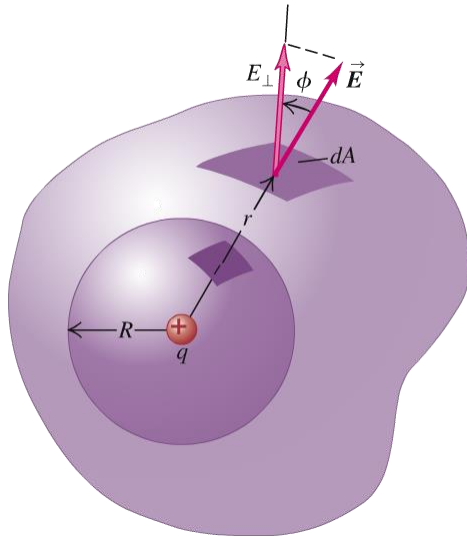


A brief review

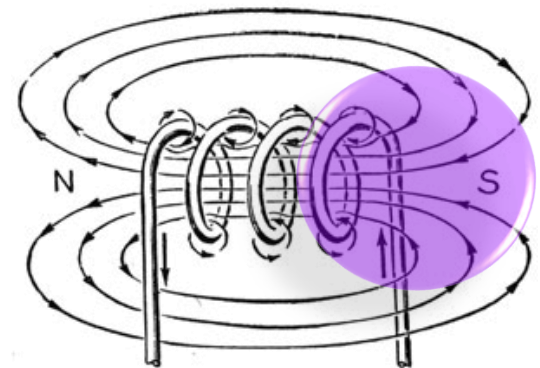
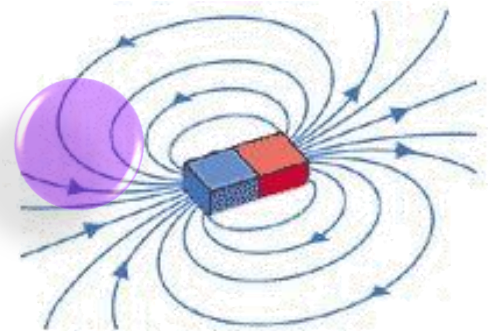
Gauss's law for $\vec{E}$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$



Gauss's law for $\vec{B}$

$$\oint \vec{B} \cdot d\vec{A} = 0$$

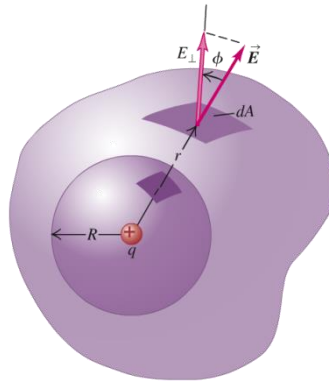


A brief review

Gauss's law for $\vec{E}$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

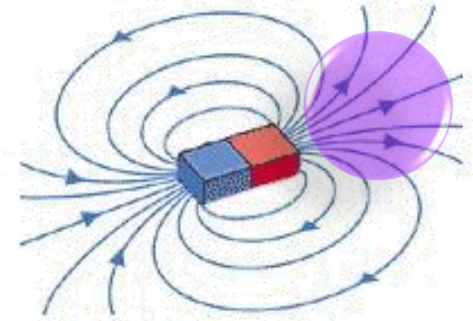
closed **surface** $\rightarrow$ area



Gauss's law for $\vec{B}$

$$\oint \vec{B} \cdot d\vec{A} = 0$$

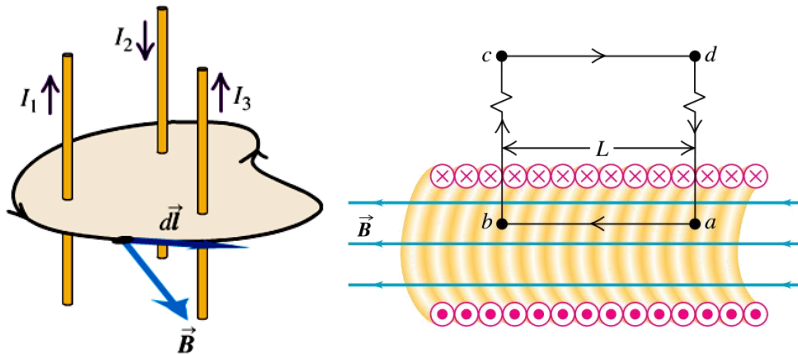
closed **surface** $\rightarrow$ area



Ampere's law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

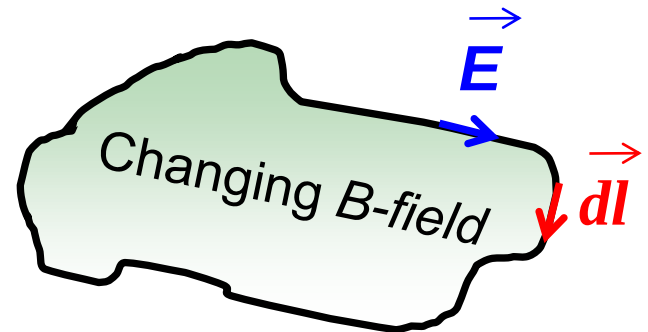
closed **loop** $\rightarrow$ length



Faraday's law

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A}$$

closed **loop** $\rightarrow$ length



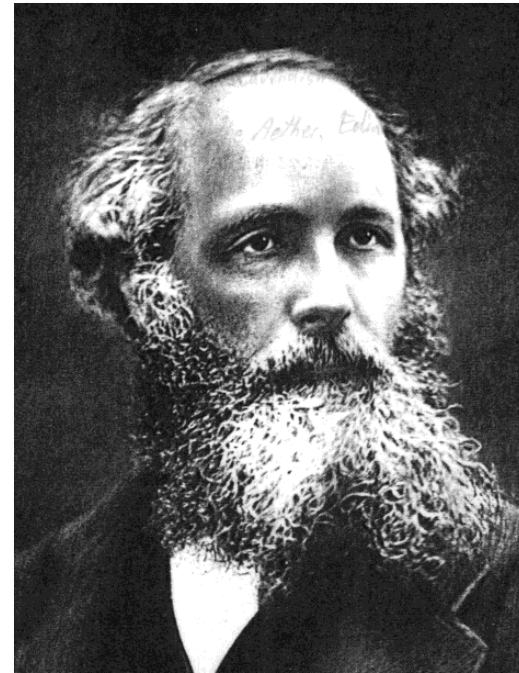
Chap.32 Maxwell's Equations

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

$$\oint \vec{B} \cdot d\vec{A} = 0$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

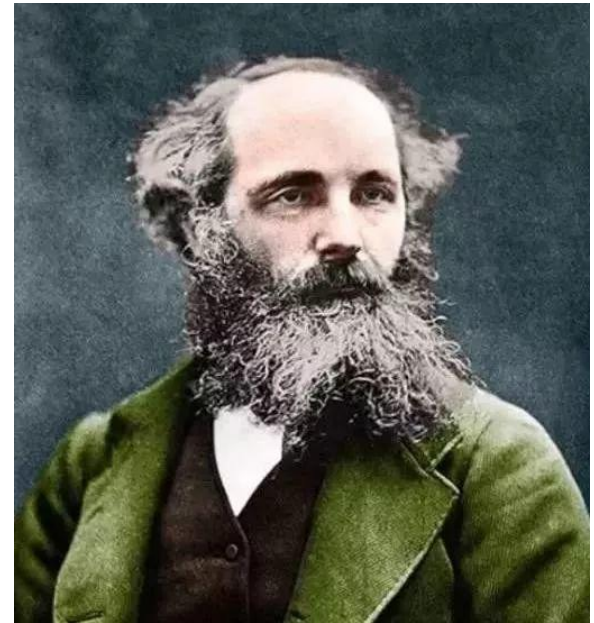


James Clerk Maxwell
(1831-1879)

Chap.32 Maxwell's Equations

James Clerk Maxwell (1831 –1879) was a Scottish scientist in the field of mathematical physics. His most notable achievement was to formulate the classical theory of electromagnetic radiation, bringing together for the first time electricity, magnetism, and light as different manifestations of the same phenomenon.

$\oiint_{\partial\Omega} \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\epsilon_0} \iiint_{\Omega} \rho dV$	$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$
$\oiint_{\partial\Omega} \mathbf{B} \cdot d\mathbf{S} = 0$	$\nabla \cdot \mathbf{B} = 0$
$\oint_{\partial\Sigma} \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \iint_{\Sigma} \mathbf{B} \cdot d\mathbf{S}$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$
$\oint_{\partial\Sigma} \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 \iint_{\Sigma} \mathbf{J} \cdot d\mathbf{S} + \mu_0 \epsilon_0 \frac{d}{dt} \iint_{\Sigma} \mathbf{E} \cdot d\mathbf{S}$	$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$



Albert Einstein: *The work of James Clerk Maxwell changed the world forever*

James Clerk Maxwell

Some symmetry concern.....

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \quad \oint \vec{B} \cdot d\vec{A} = 0$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \quad \oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

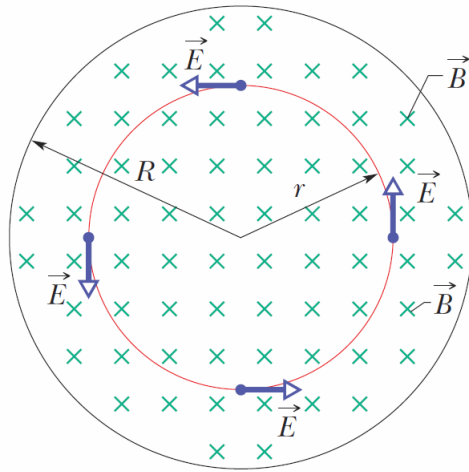
$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

A changing magnetic flux induces an E-field along a closed loop

By symmetry, can a changing electric flux induces a B-field along a closed loop?

Something like: $\oint \vec{B} \cdot d\vec{l} = (\sim!@\$ \% \wedge \& *) \frac{d\Phi_E}{dt} ?$

Induction Laws

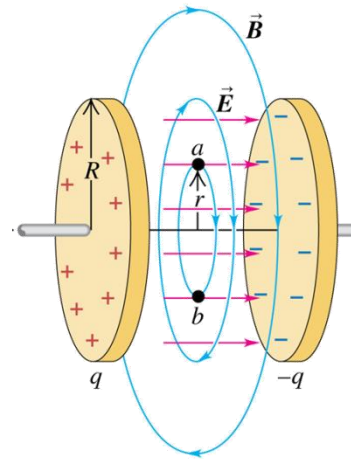


Increasing B field

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

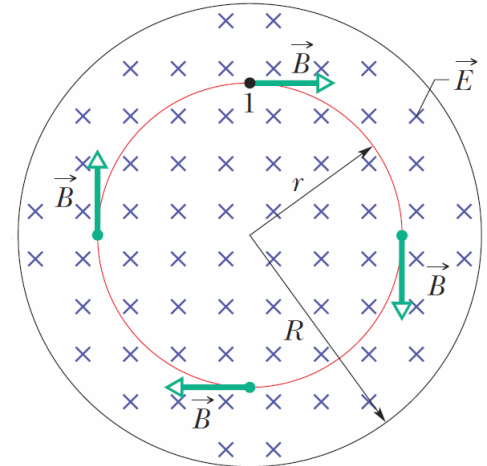
$$E 2\pi r = -\frac{dB}{dt} \pi r^2$$

$$\Rightarrow E = -\frac{1}{2} \frac{dB}{dt} r$$



Increasing E field

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$



Direction of the E is *counterclockwise*!

Direction of the E is *clockwise*!

Ampere-Maxwell's Law and Displacement Current

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_{\text{enc}}$$

Ampere-Maxwell Law:
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_{\text{enc}} + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

It is obvious that $\varepsilon_0 d\Phi_E/dt$ has the same unit as current. In fact, we can define this term as a “displacement current”,

$$i_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

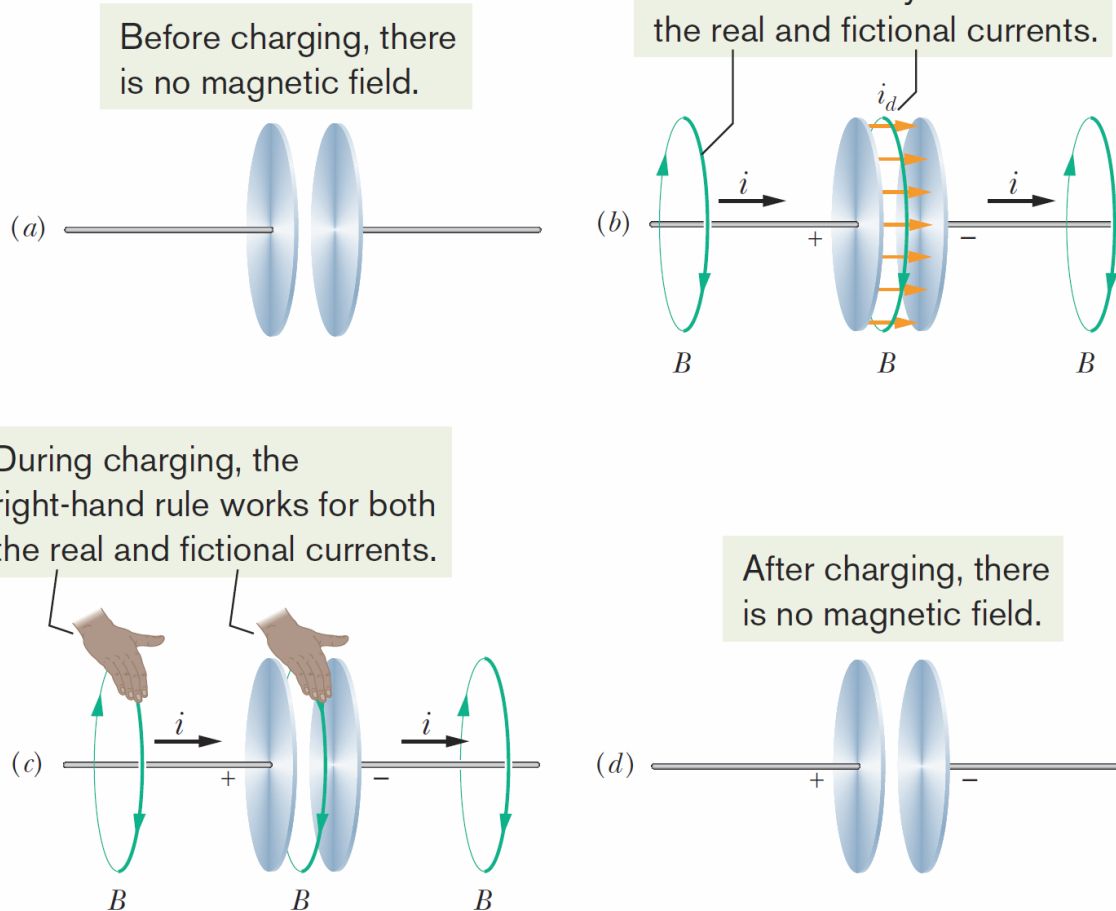
such that the Ampere-Maxwell's law can be written as

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i_{\text{enc}} + i_d)$$

Displacement current is not due to motion of electric charges, so it is a “fictional” current!

More on Displacement Current

We can get a sense of the displacement current in the configuration of a charging capacitor.



$$\begin{aligned}
 i &= \frac{dq}{dt} \\
 &= \frac{d(Cv)}{dt} \\
 &= C \frac{d(Ed)}{dt} \\
 &= \frac{\epsilon_0 A}{d} d \frac{dE}{dt} \\
 &= \epsilon_0 \frac{d(AE)}{dt} \\
 &= \epsilon_0 \frac{d\Phi_E}{dt} \\
 &= i_d
 \end{aligned}$$

Maxwell's equations*

* Assume there is no dielectric or magnetic materials present

$$\text{Gauss' law for electricity} \quad \oint \vec{E} \cdot d\vec{A} = q_{\text{enc}}/\epsilon_0$$

Net electric flux

Net enclosed electric charge

$$\text{Gauss' law for magnetism} \quad \oint \vec{B} \cdot d\vec{A} = 0$$

Net magnetic flux

Net enclosed magnetic charge

$$\text{Faraday's law} \quad \oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$$

Induced E-field

Changing magnetic flux

$$\text{Ampere-Maxwell law} \quad \oint \vec{B} \cdot d\vec{s} = \mu_0\epsilon_0 \frac{d\Phi_E}{dt} + \mu_0 i_{\text{enc}}$$

Induced B-field

Changing electric flux and current

Different Forms of Maxwell's Equations

Integral Form

$$\left\{ \begin{array}{l} \oint_{\mathbf{A}=\partial V} \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\varepsilon_0} \\ \oint_{\mathbf{A}=\partial V} \mathbf{B} \cdot d\mathbf{A} = 0 \\ \oint_{\mathbf{l}=\partial \mathbf{A}} \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_{\mathbf{A}} \mathbf{B} \cdot d\mathbf{A} \\ \oint_{\mathbf{l}=\partial \mathbf{A}} \mathbf{B} \cdot d\mathbf{l} = \mu_0 \left(\int_{\mathbf{A}} \mathbf{J} \cdot d\mathbf{A} + \varepsilon_0 \frac{d}{dt} \int_{\mathbf{A}} \mathbf{E} \cdot d\mathbf{A} \right) \end{array} \right.$$

Differential Form

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \end{array} \right.$$

Even more elegant form

$$\partial^\nu F_{\mu\nu} = \mu_0 \vec{J}_\mu$$

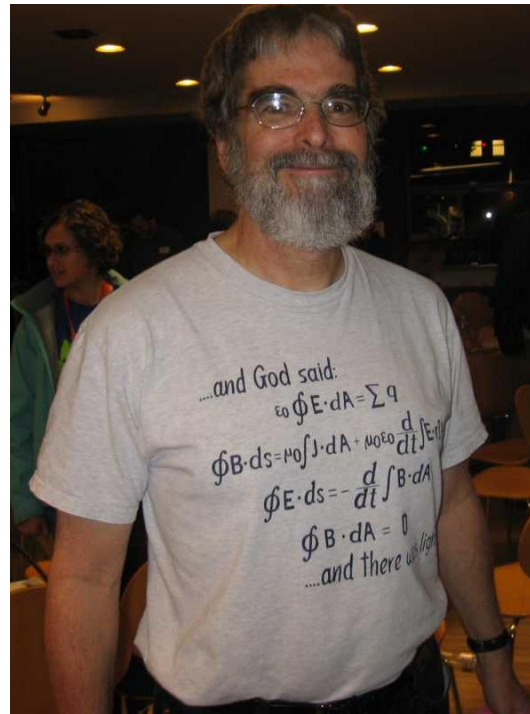
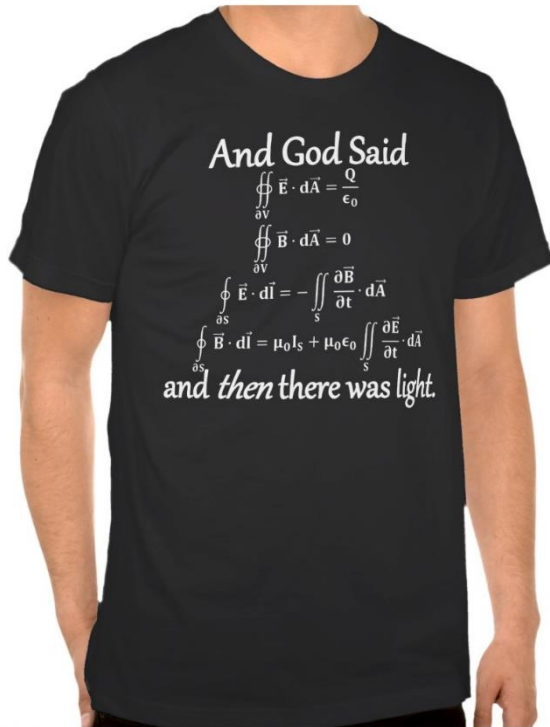
$$\partial^\lambda F_{\mu\nu} + \partial^\mu F_{\nu\lambda} + \partial^\nu F_{\lambda\mu} = 0$$

矢量微分、矢量积分、张量、四维矢势、外微分。。。。。

Beauty of Physics

Some of the most beautiful equations:

1. Maxwell's equations;
2. $\mathbf{F} = m\mathbf{a}$;
3. Einstein's equation of general relativity; ...

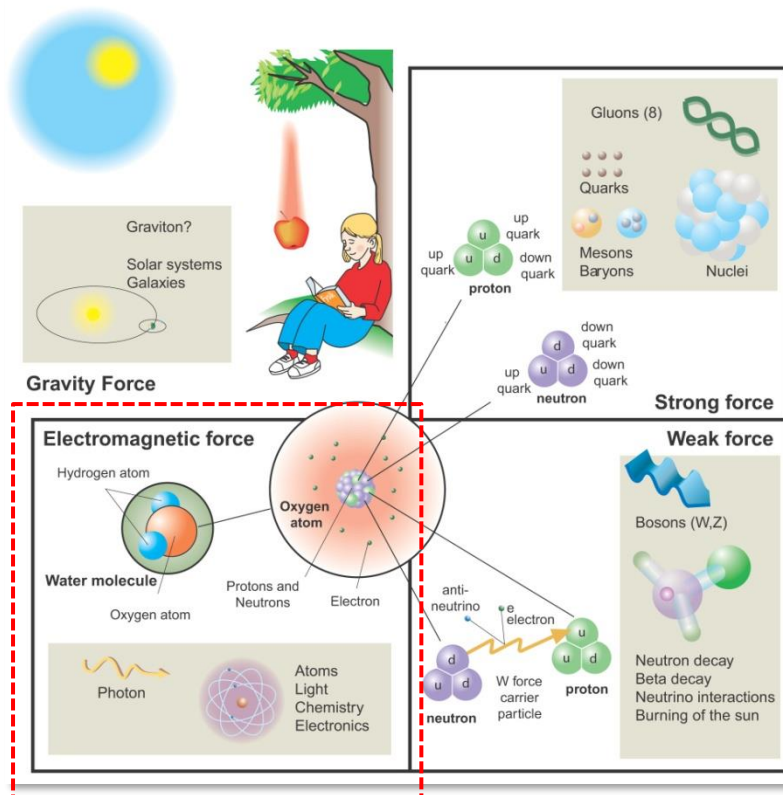


From a long view of the history of mankind — seen from, say, ten thousand years from now — there can be little doubt that **the most significant event of the 19th century will be judged as Maxwell's discovery of the laws of electrodynamics.** The American Civil War will pale into provincial insignificance in comparison with this important scientific event of the same decade.

----Richard Feynman

Maxwell's Equations Rule the Universe Almost

Different forces: friction, normal force, tension, spring force, buoyancy, gravitational force...



Fundamental Forces				
Strong	Force which holds nucleus together	Strength 1	Range (m) 10^{-15} (diameter of a medium sized nucleus)	Particle gluons, π (nucleons)
Electro-magnetic		Strength $\frac{1}{137}$	Range (m) Infinite	Particle photon mass = 0 spin = 1
Weak	neutrino interaction induces beta decay	Strength 10^{-6}	Range (m) 10^{-18} (0.1% of the diameter of a proton)	Intermediate vector bosons W^+ , W^- , Z_0 , mass > 80 GeV spin = 1
Gravity		Strength 6×10^{-39}	Range (m) Infinite	Particle graviton ? mass = 0 spin = 2

Electromagnetic (EM) waves

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Changing **B** $\rightarrow$ **E**

$$\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{J}$$

Changing **E** $\rightarrow$ **B**

In vacuum where there is no charge and current:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \nabla \cdot \vec{E} = 0 \quad \nabla \cdot \vec{B} = 0$$

A possible solution which fits all 4 Maxwell's equation :

$$\vec{E} = \vec{E}_0 \cos(kx - \omega t)$$

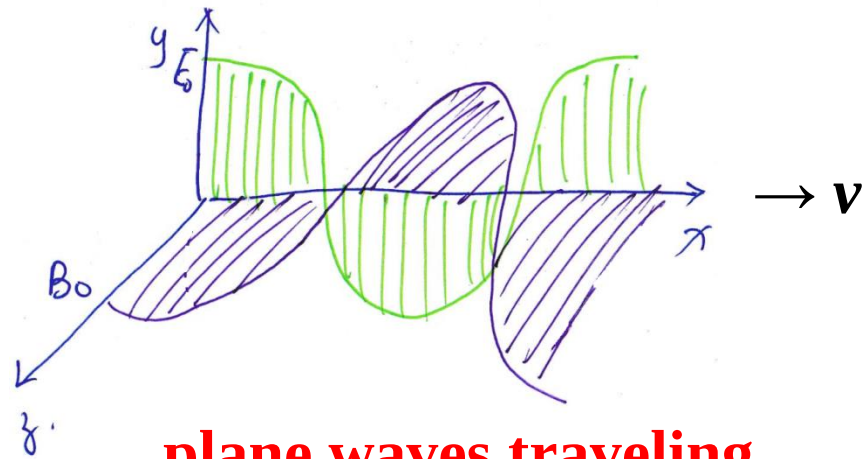
$$\vec{B} = \vec{B}_0 \cos(kx - \omega t)$$

$$\omega / k = v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \quad 2\pi / \lambda = k$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ F/A}^2$$

$$v = 2.999 \times 10^8 \text{ m/s}$$

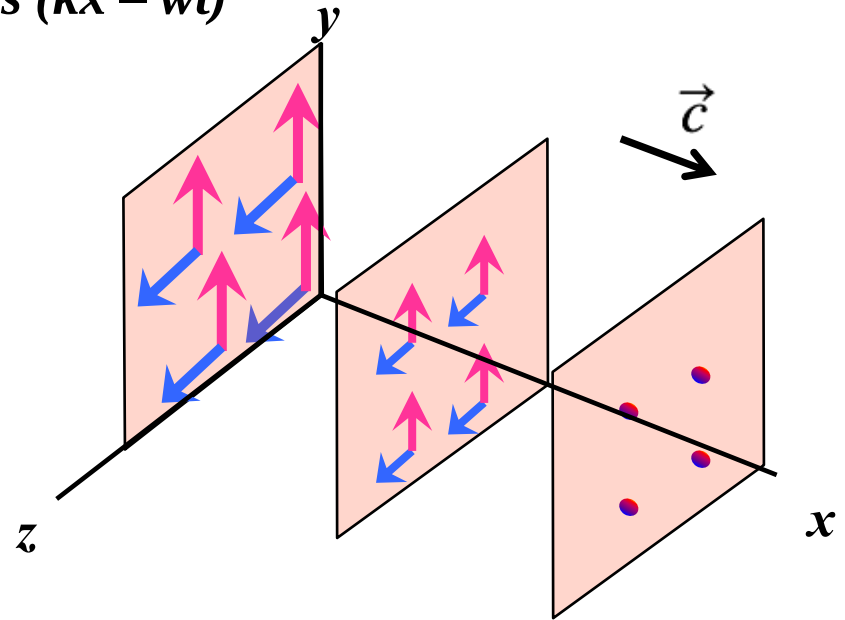
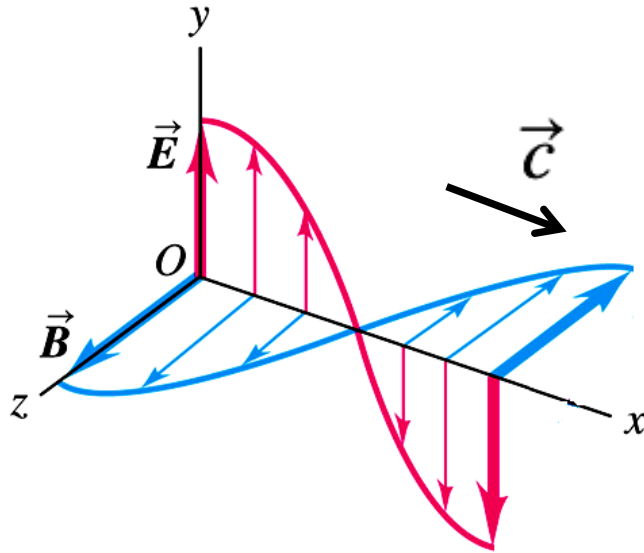


**plane waves traveling
at the speed of light!!**

Electromagnetic Waves

$$\vec{E} = \vec{E}_0 \cos(kx - \omega t)$$

$$\vec{B} = \vec{B}_0 \cos(kx - \omega t)$$



$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\vec{E} \perp \vec{c}$$

$$\vec{B} \perp \vec{c}$$

$$\vec{E} \perp \vec{B}$$

$$\vec{E} \text{ \& \; } \vec{B} \text{ in phase}$$

$$\vec{E} \times \vec{B} = \vec{c}$$

$$\frac{E_0}{B_0} = c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Electromagnetic Waves

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

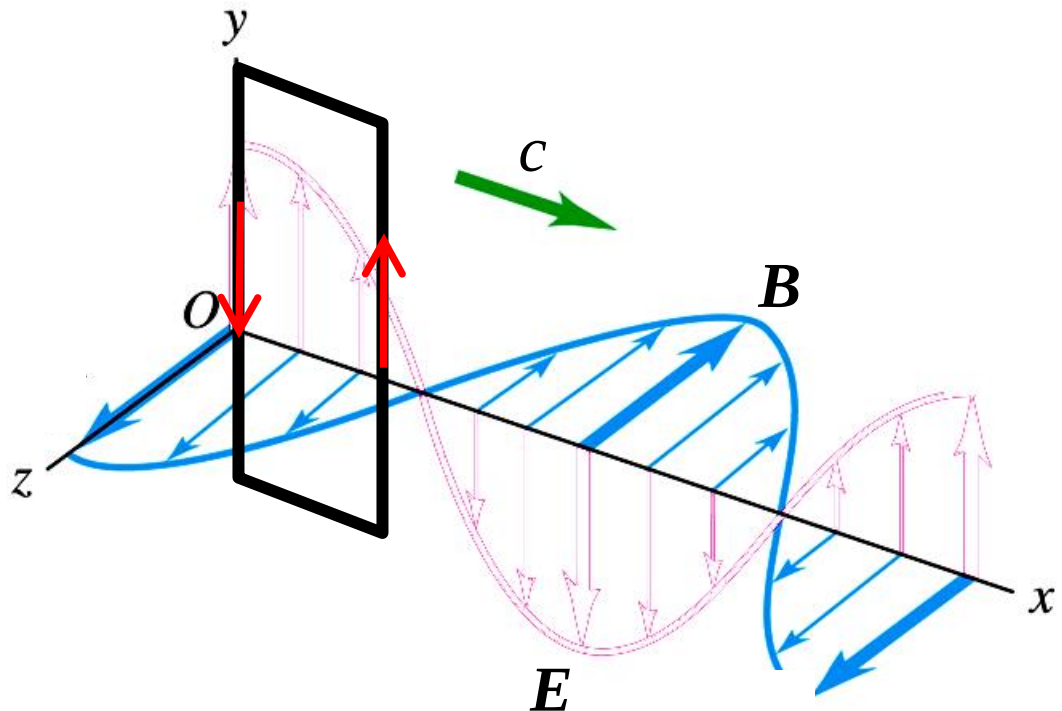
$$\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\vec{E} = \vec{E}_0 \cos(kx - \omega t)$$

$$\vec{B} = \vec{B}_0 \cos(kx - \omega t)$$



Electromagnetic Waves

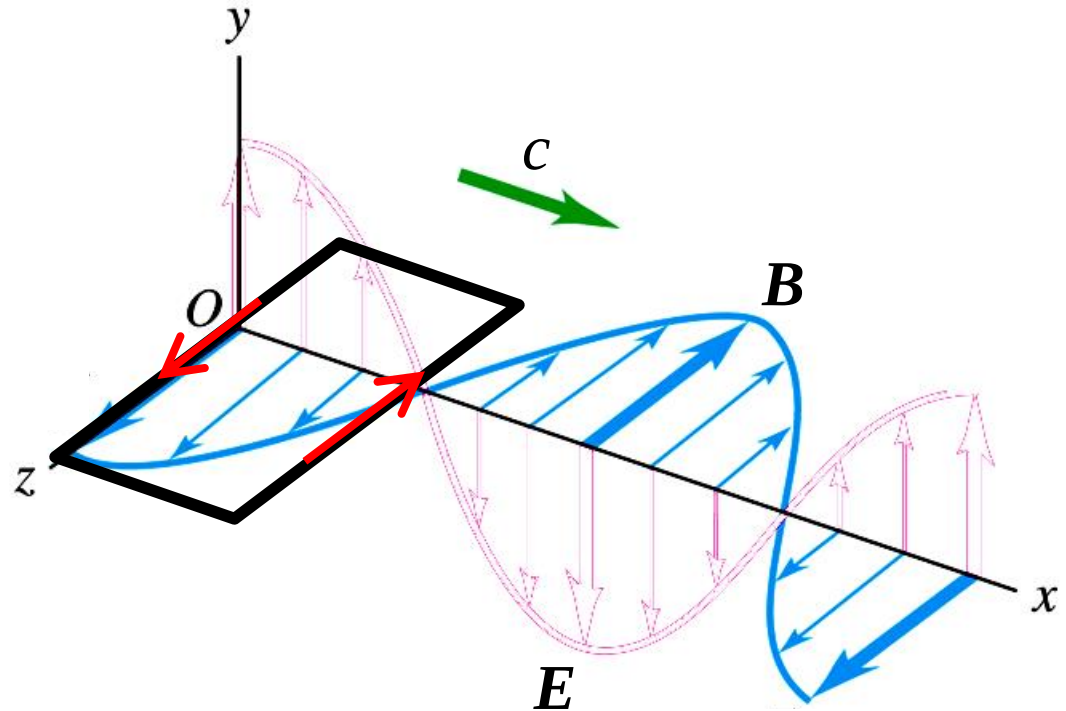
$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\vec{E} = \vec{E}_0 \cos(kx - \omega t)$$
$$\vec{B} = \vec{B}_0 \cos(kx - \omega t)$$



Derivation of the Wave Equation

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \nabla \cdot \vec{E} = 0 \quad \nabla \cdot \vec{B} = 0$$

$$\begin{aligned} \nabla \times (\nabla \times \vec{E}) &= \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\nabla^2 \vec{E} = \nabla \times \left(-\frac{\partial \vec{B}}{\partial t}\right) \\ &= -\frac{\partial(\nabla \times \vec{B})}{\partial t} = -\frac{\partial(\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t})}{\partial t} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \end{aligned}$$

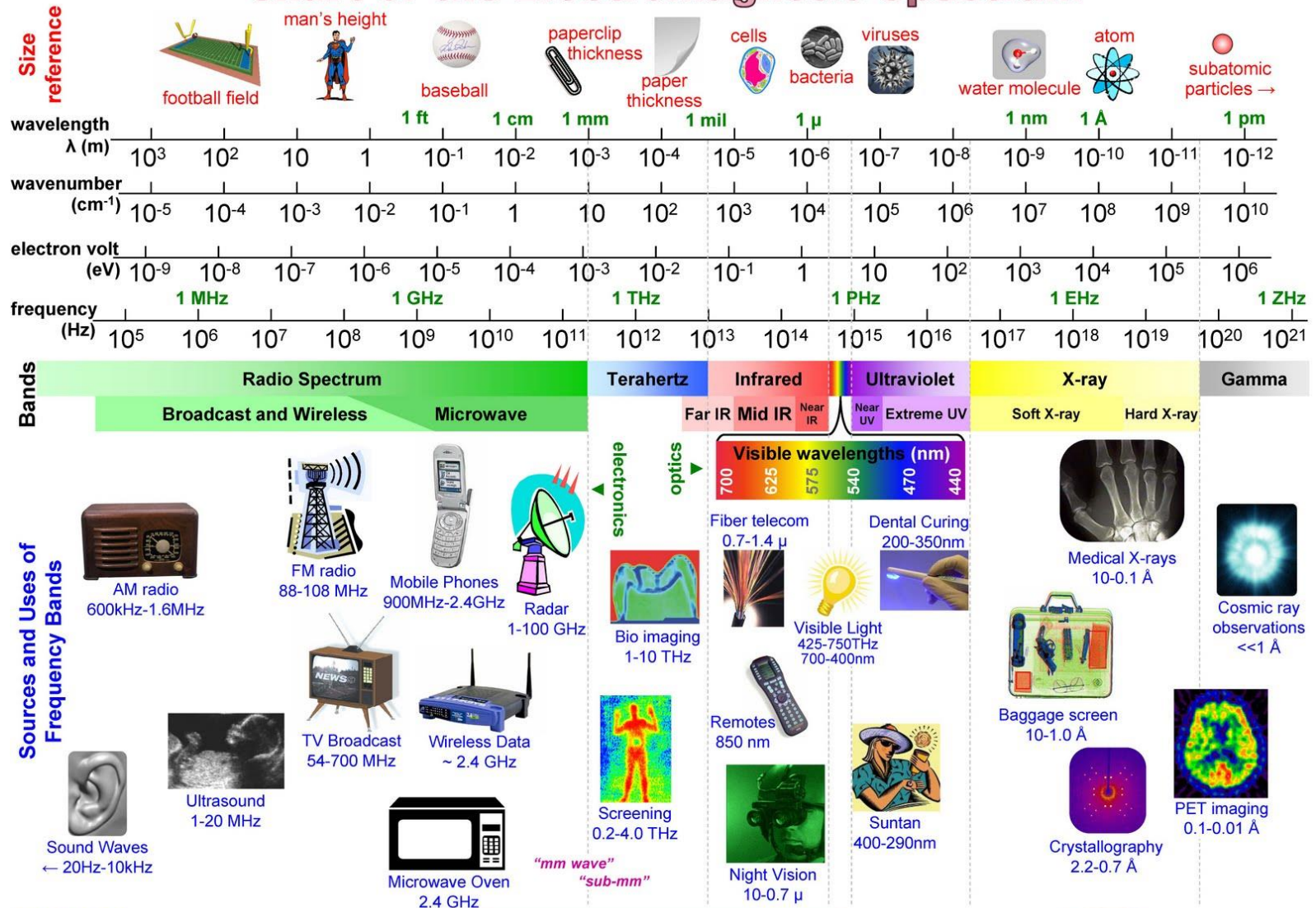
$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{What is this? Wave with a speed of } \frac{1}{\sqrt{\mu_0 \epsilon_0}} !$$

$$\nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

A non-高大上 derivation of the wave equation.....

A Universe Filled with EM Waves

Chart of the Electromagnetic Spectrum

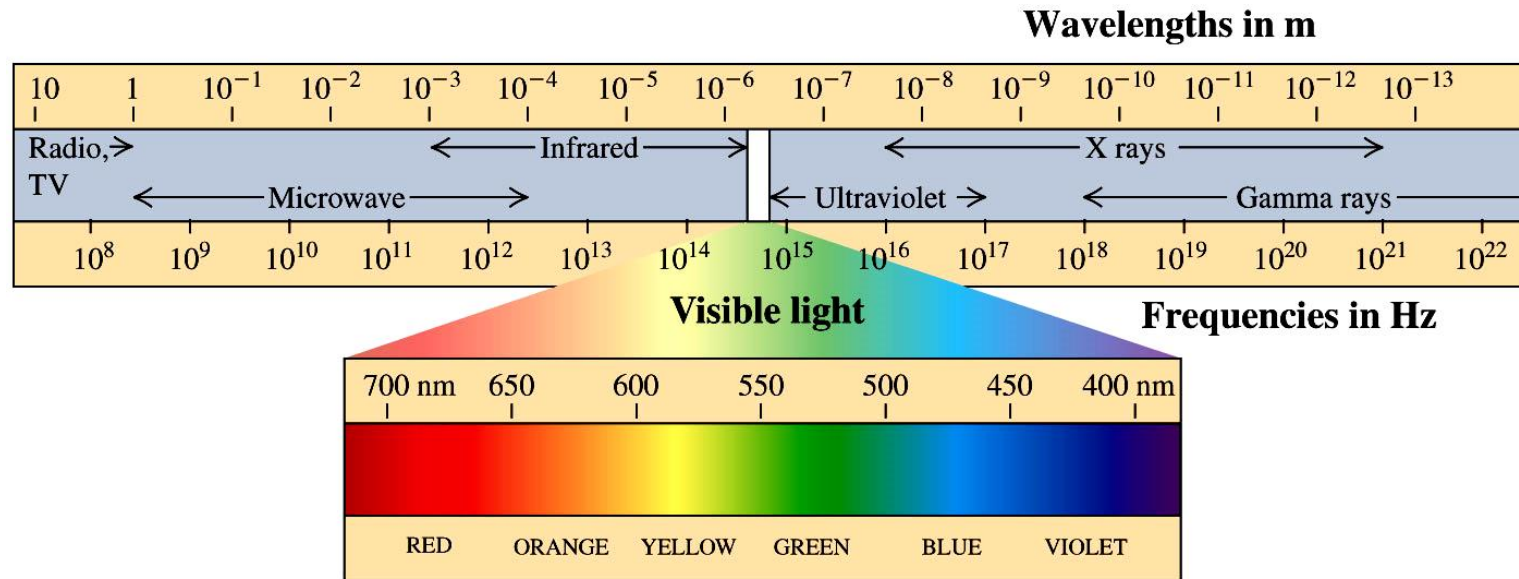


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$$\lambda = 3 \times 10^8 / \text{freq} = 1 / (\text{wn} \times 100) = 1.24 \times 10^{-6} / \text{eV}$$

SURA Southeastern Universities
 Research Association ®

Visible Lights



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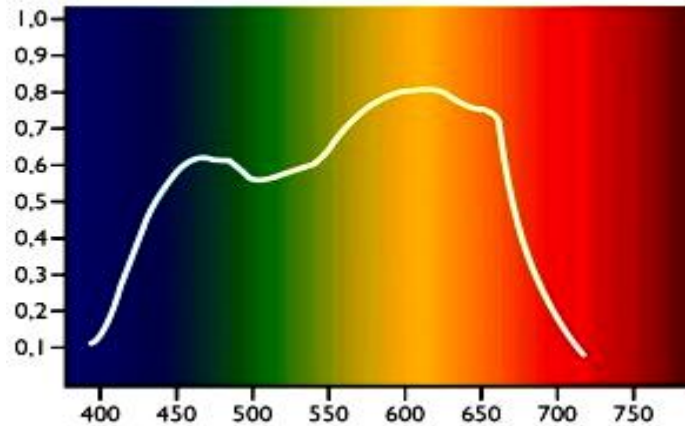
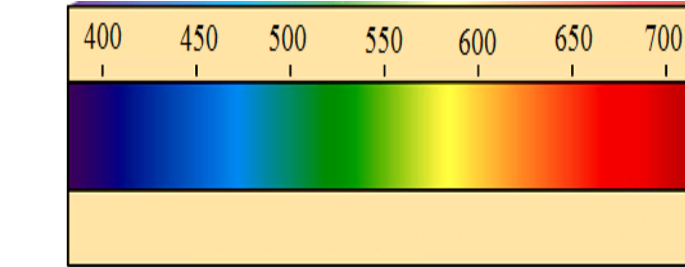
Wavelengths of Visible Light

380–450 nm	Violet
450–495 nm	Blue
495–570 nm	Green
570–590 nm	Yellow
590–620 nm	Orange
620–750 nm	Red

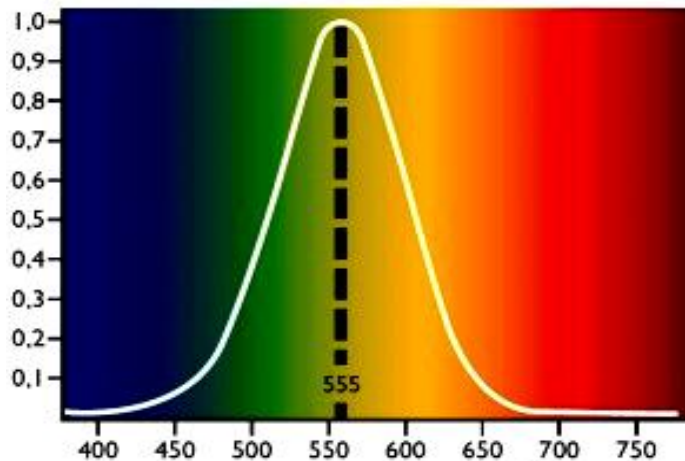


Visible Lights

Plant sensitivity curve for photosynthesis

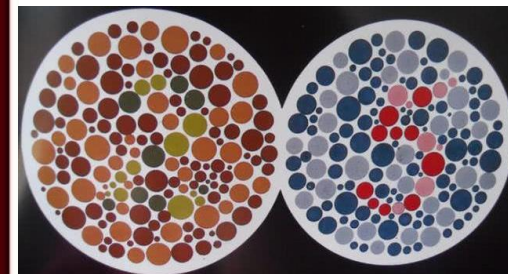
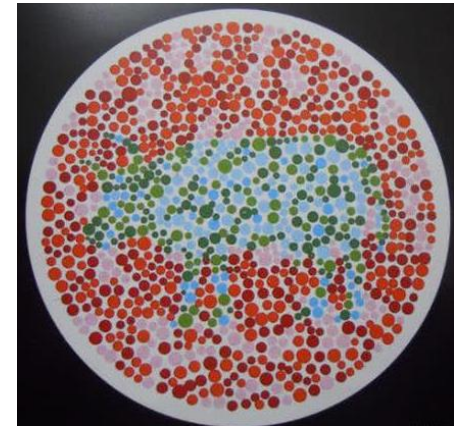


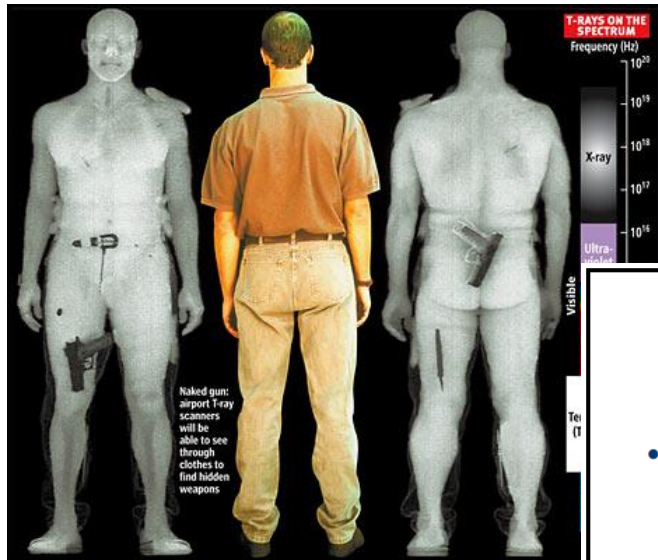
Human eye sensitivity to colors



Women have a better differentiating power concerning colors with low wavelengths than men.

Who can see this?



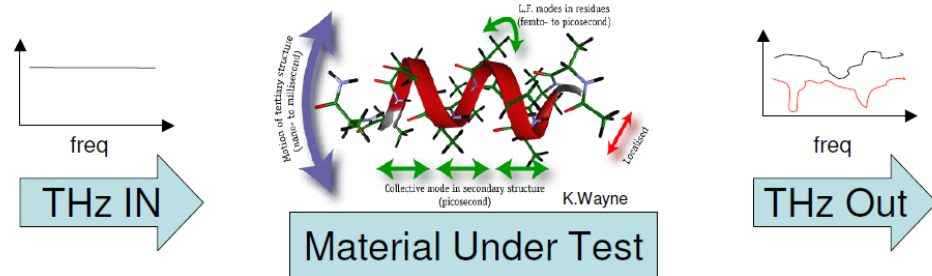


Terahertz spectroscopy

Terahertz scanner in the airport

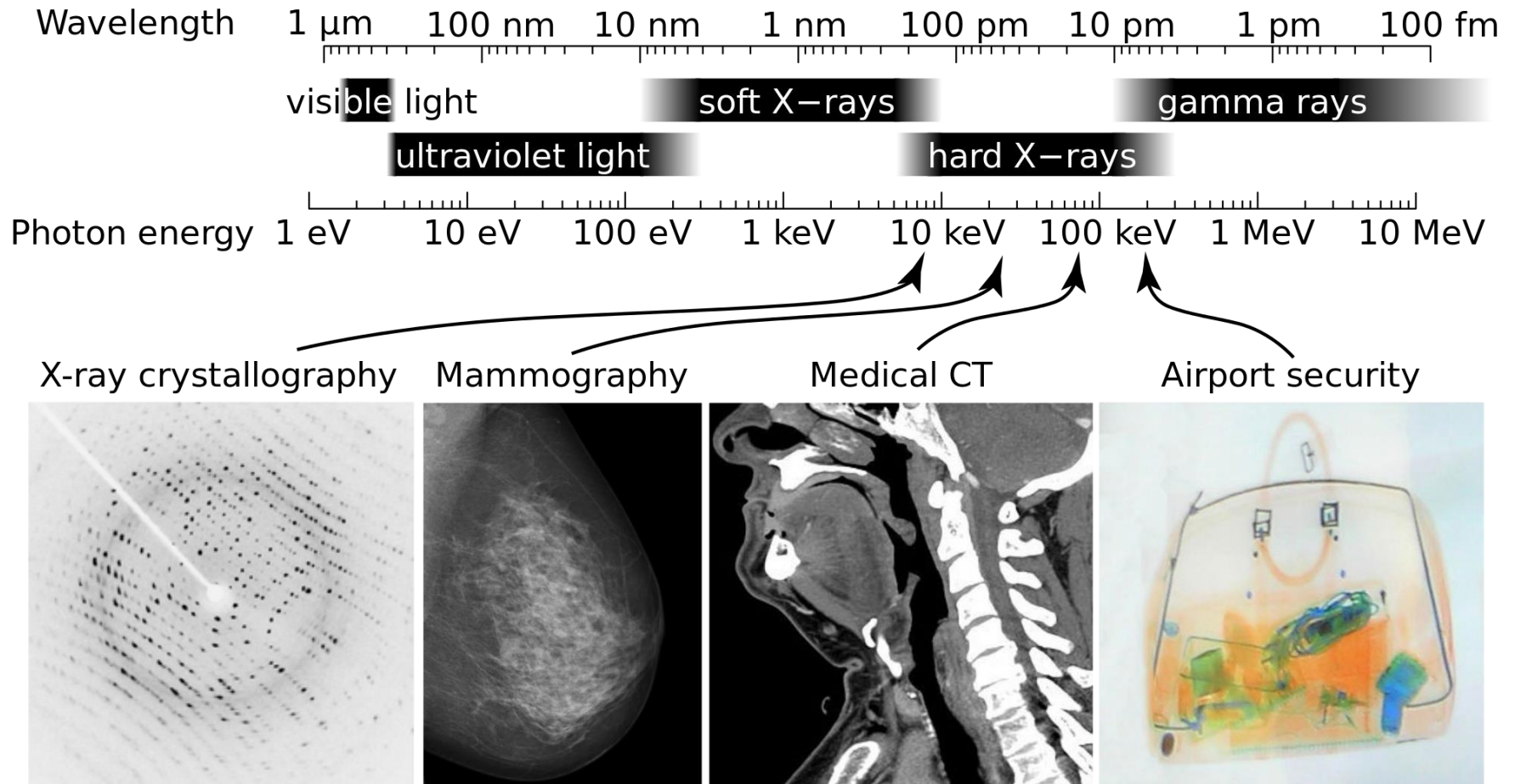
How can we use terahertz waves ?

- Spectroscopy / Absorption Spectra

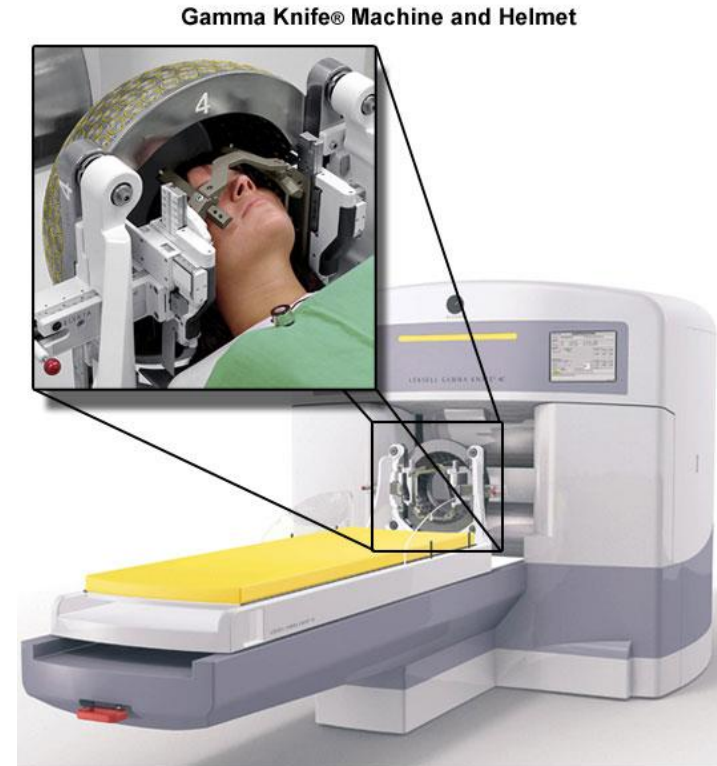
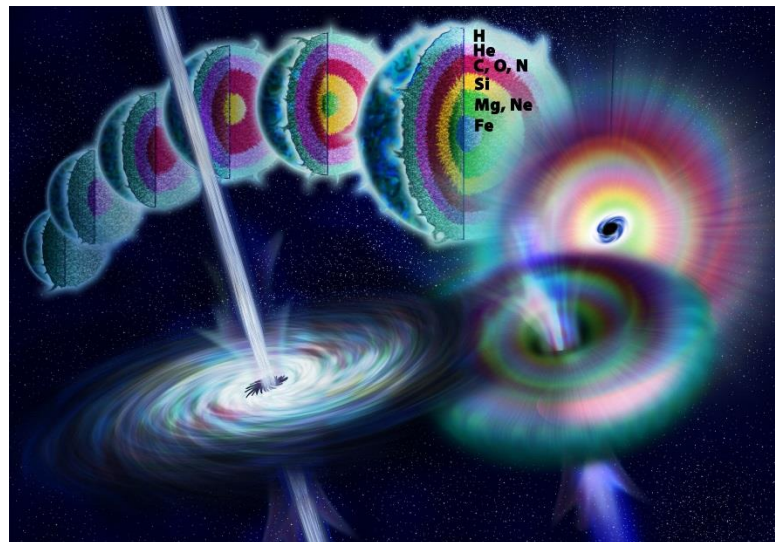
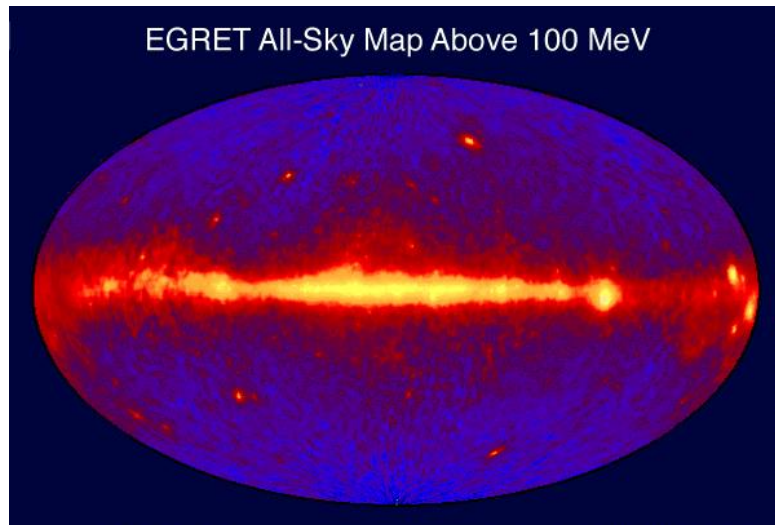


- Organic molecules exhibit strong absorption from GHz to THz through rotational and vibration transitions providing fingerprints in the THz band.
- THz material 'signatures'
- The crystalline or physical structure of materials provides further terahertz absorptions.

X-ray



Gamma-ray



Images Courtesy of Elekta, Inc.

***“Nature and nature’s law lay hid in night,
God said: Let Maxwell be!
and all was light.”***

Modified based on **Alexandra Pope**

Energy transfer in EM waves

Energy stored in electric field per unit volume

$$u_E = \frac{1}{2} \epsilon_0 E^2 \quad (\text{J/m}^3)$$

Energy stored in magnetic field per unit volume

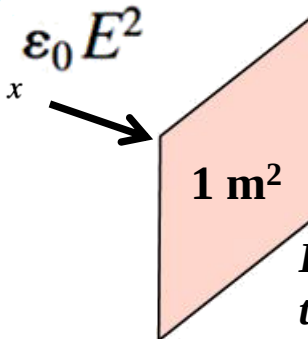
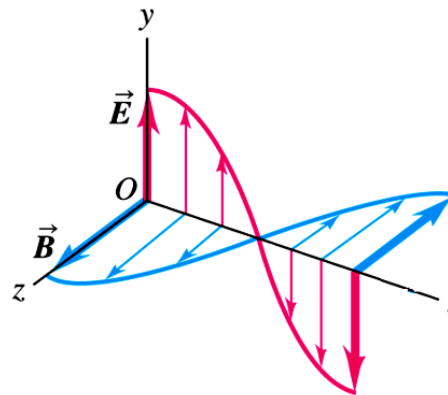
$$u_B = \frac{B^2}{2\mu_0}$$

In traveling EM waves:

$$\begin{aligned}\vec{E} &= \vec{E}_0 \cos(kx - \omega t) \\ \vec{B} &= \vec{B}_0 \cos(kx - \omega t)\end{aligned}$$

$$\frac{E_0}{B_0} = c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\begin{aligned}\therefore u_B &= \frac{B^2}{2\mu_0} \\ &= \frac{1}{2\mu_0} \frac{E^2}{c^2} \\ &= \frac{E^2}{2\mu_0} \mu_0 \epsilon_0 \\ &= \frac{1}{2} \epsilon_0 E^2 \\ &= u_E !\end{aligned}$$



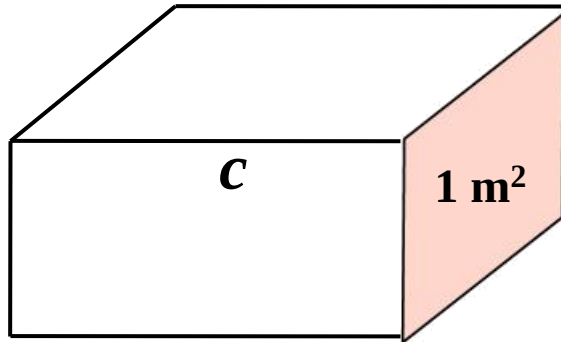
How much energy is transfer through this area in every second?

$$u_{total} = \epsilon_0 E^2 = \epsilon_0 E B c$$

Energy transfer in EM waves

How much energy flows through $1 \text{ m}^2 (\perp)$ every second?
i.e. *energy transfer per unit time per unit cross-sectional area*
or *power per unit area*

Light travel a distance of c in 1 sec



Volume of the box: $c \text{ m}^3$

In every second, energy inside the box
will pass through this area of 1 m^2

$\therefore$ Energy comes out per unit second (energy flux):

$$u_{\text{total}} c = \epsilon_0 E^2 c = \epsilon_0 E B = \frac{EB}{\mu_0} \text{ J/(m}^2 \text{ sec)}$$

Introducing the Poynting vector:

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} \quad \begin{array}{l} \text{J/(m}^2 \text{ sec)} \\ \text{W/m}^2 \end{array}$$

Energy transfer in EM waves

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

Since $\mathbf{E}$ and $\mathbf{B}$ are changing, it is more meaningful to use its average value.

So we have :

$$S_{avg} = \frac{E_0 B_0}{2 \mu_0} = \frac{E_0^2}{2 \mu_0 c} \quad ?$$

$$\begin{aligned}\vec{E} &= \vec{E}_0 \cos(kx - \omega t) \\ \vec{B} &= \vec{B}_0 \cos(kx - \omega t)\end{aligned}$$

and we call this the **intensity** I of the EM waves

$$\text{and since } E_{rms} = \frac{E_0}{\sqrt{2}}$$

$$\text{we can write } I = S_{avg} = \frac{E_{rms}^2}{\mu_0 c} \quad (W/m^2) \quad \left(\frac{\text{power}}{\text{area}} \right)_{avg}$$

change of **intensity** I with distance for a point source :

$$\frac{\text{average power}}{4\pi r^2} \propto \frac{1}{r^2}$$

Review and Summary

Review & Summary

Gauss' Law for Magnetic Fields The simplest magnetic structures are magnetic dipoles. Magnetic monopoles do not exist (as far as we know). **Gauss' law** for magnetic fields,

$$\Phi_B = \oint \vec{B} \cdot d\vec{A} = 0, \quad (32-1)$$

states that the net magnetic flux through any (closed) Gaussian surface is zero. It implies that magnetic monopoles do not exist.

Maxwell's Extension of Ampere's Law A changing electric flux induces a magnetic field $\vec{B}$. Maxwell's law,

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} \quad (\text{Maxwell's law of induction}), \quad (32-3)$$

relates the magnetic field induced along a closed loop to the changing electric flux Φ_E through the loop. Ampere's law,

$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{\text{enc}}$ (Eq. 32-4), gives the magnetic field generated by a current i_{enc} encircled by a closed loop. Maxwell's law and Ampere's law can be written as the single equation

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} + \mu_0 i_{\text{enc}} \quad (\text{Ampere-Maxwell law}). \quad (32-5)$$

Displacement Current We define the fictitious *displacement current* due to a changing electric field as

$$i_d = \epsilon_0 \frac{d\Phi_E}{dt}. \quad (32-10)$$

Equation 32-5 then becomes

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{d,\text{enc}} + \mu_0 i_{\text{enc}} \quad (\text{Ampere-Maxwell law}), \quad (32-11)$$

where $i_{d,\text{enc}}$ is the displacement current encircled by the integration

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截图(Alt + A)

loop. The idea of a displacement current allows us to retain the notion of continuity of current through a capacitor. However, displacement current is *not* a transfer of charge.

Maxwell's Equations Maxwell's equations, displayed in Table 32-1, summarize electromagnetism and form its foundation, including optics.

Orbital angular momentum is quantized and can have only measured values given by

$$L_{\text{orb},z} = m_\ell \frac{h}{2\pi},$$

for $m_\ell = 0, \pm 1, \pm 2, \dots, \pm (\text{limit})$. (32-29)

The associated magnetic dipole moment is given by